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Choosing Optimum Number of Fracturing Sleeves in Multi-Entry Cemented MSF Completions Advances Operational Efficiency and Reduces Costs in Gas Wells

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Abstract

Cemented multi-stage fracturing (CMSF) completion was the first of its kind to be implemented in a deep tight gas well. A single-entry (SE) system was first introduced then an upgrade was developed with a multi-entry (ME) system to stimulate gas reservoirs in multi stages. This paper will illustrate the operational efficiency gain from the ME completions and will compare stages and number of fracturing sleeves in terms of injection efficiency and production gains.

CMSF systems are alternative to open-hole MSF (OHMSF) systems and plug-and-perf (P&P) systems. All are designed for multi-stage fracturing in targeted laterals to maximize production while reducing the drawdown pressure. CMSF systems are run similar to OHMSF systems with fracturing sleeves/ports but without OH packers to isolate between stages and cement is pumped between stages for isolation. The completion toe opens up by applying a certain pressure and the first stage can be pumped through it. Other stages open through ball drops to activate fracturing sleeves. The number of entries is studied against injection efficiency and production gains.

Operational efficiency was improved in ME systems compared to SE systems. One ME well had 10 stimulation stages pumped per design. The toe stage had two entries while remaining 9 stages had three entries per stage. Previously, 9 stages were planned in the SE system and 7 stages were pumped per design while 2 stages were skipped due to low injectivity. Both systems capitalized on cement for isolation between stages and not using OH packers that could fail and cause communication between stages. Additionally, the ME system capitalized to pump all planned stages as more flow area is available. Gas rate is gained as more effective stages are pumped and cost is reduced as less lost time is encountered. It is worth to mention that drilling or completion running issues are significantly reduced for running these CMSF systems even in tight developments compared to OHMSF. Gas wells can even run it without a mechanical caliper. Finally, there should be an optimum number of stages or entries per stage to avoid low injectivity or over design. This paper will show a well that needed more entries and will show another that needed less.

Stuck chances are reduced while running the CMSF components as the completion OD is reduced compared to the OHMSF. Fracturing sleeves are placed with the help of petrophysical data and AI caliper

log while mechanical calipers can be no longer needed. Finally, production gain is realized, cost is saved, and wellbore exposure time is reduced.

Introduction

Multi-stage fracturing (MSF) in gas wells were introduced for some decades now with the intention to enhance productivity and sustainability of production. This is compared to non-stimulated wells whether vertical or horizontal wells that could have a decline in production in a few months or years depending on the reservoir quality that they are drilled in. Gas wells drilled in the sweetest spots could certainly sustain some production even if hydraulic fracturing is not used. However, a common preference would be to get higher initial gas rates and certainly more sustainable production as much as possible. Stimulating wells as soon as possible is certainly recommended to deliver on this preference whether in vertical or horizontal wells.

Horizontal gas wells with extended reach then completed with MSF are aimed mainly to lower the drawdown pressure from multi entries and maintain healthy reservoirs. They also aim to reduce the carbon footprint that is by avoiding or at least minimizing well interventions post tie-in to gas plants. Also, wells drilled away from sweet reservoir spots or across tight developments certainly need horizontal wells drilled in the minimum stress direction (et al. Rahim 2011), higher pay contact and more stimulated stages to ensure a longer well life. MSF in horizontal wells is a necessity more than a luxury in these tight developments or low permeability reservoirs. The associated stimulation costs are easily justified as production gains are achieved from an effective stimulation strategy. Some strategies could require OHMSF while others could require CMSF.

OHMSF completion was first used with single-entry (SE) ports to stimulate desired number of stages then it was upgraded to multi-entry (ME) ports while using OH packers between stages. The ME ports are used to maximize the stimulation contact as some reservoir or stages could only need an acid matrix/wash job rather than a hydraulic fracture to produce. If the ME ports are expected to create ME fractures, constrictor or mini packers were introduced to be placed between ports but they normally do not allow the completion to rotate while running in hole. Anyone would think that the more stages the merrier as more production should be achieved. However, overdesigning OHMSF completions in terms of stages, ports, or using costly rotatable packers in a gas well could just lead to a higher cost rather than gaining additional production whether it was in SE or ME systems. On the other hand, under-designed lower completion could lead to missed opportunity or opportunities, shorter well life or advancement of workover costs.

CMSF completion was later introduced in gas wells with more flexibility in deployment than OHMSF completions like the ability to rotate while running the lower completion. Also, cement will be used between stages instead of OH packers that cannot rotate or instead of costly OH packers that could rotate. CMSF was also introduced with a SE system then was upgraded to a ME system. Similar to OHMSF, there could be overdesigned or under-designed completions. Overdesigned completions like using more fracturing sleeves than needed where under-designed completions like having missed opportunity/ies, additional stimulation stage or stages. Also, injection issue could be encountered, when using a SE stage and not a ME stage, that could lead to high-pressure coiled tubing (HPCT) operation to enhance injection that will translate to inefficient stimulation job.

It was noted that in all completions it is important to reduce the exposure time of any drilled reservoir/formation whether to improve operational efficiency, reduce environmental risks, or to not allow the hole condition to deteriorate over time. Knowing the hole condition (size) and keeping the hole intact while deploying OHMSF or CMSF completions are extremely important for smooth and successful stimulation. It was also noted that CMSF completions can even rely on synthetic/AI caliper logs that is derived from Logging While Drilling (LWD) more than OHMSF completions but it is believed that with time, there could be more applications of AI calipers even in OHMSF completions mainly to reduce the exposure time of drilled holes as much as possible.

Multi-Stage Fracturing (MSF) Completions with Ball-drop Systems

It is well-known in the industry why MSF completions with ball-drop systems are used that is to facilitate stimulation of stages much faster than the plug-and-perf (P&P) technique that is used in cemented liners. Also, the use of a radioactive source can be eliminated that is used to correlate logs in a cemented liner (CL) completions. Moreover, perforating guns can be avoided that could be an issue logistically or operationally.

Once a gas well is completed with any kind of completion that requires injection or stimulation, a toe initiator valve (TIV) as illustrated in Figure 1 below is used so that stimulation can commence after a certain injection rate is achieved. Horizontal wells need TIVs more than vertical wells as cleaning wellbores in horizontals is more challenging using HPCT that could be costly and should be avoided to advance operational efficiency whether in a rig or in a rigless site. Rigs are normally released post opening the toe of the completion then wells are handled to the stimulation crew. Acid or proppant fracturing should then be needed in tight gas developments to enhance or establish production paths. MSF completions with ball-drop systems do not need perforations and can inject directly to the toe as the first stage or the toe can be used for cleaning the completion before dropping the delicate-in-size balls to isolate the toe then stimulate the next stage. Subsequent number of stages as desired are stimulated similarly by dropping different-size balls and each size will isolate a certain stage and open ports or fracturing sleeves for the next stage until all desired stages are stimulated.

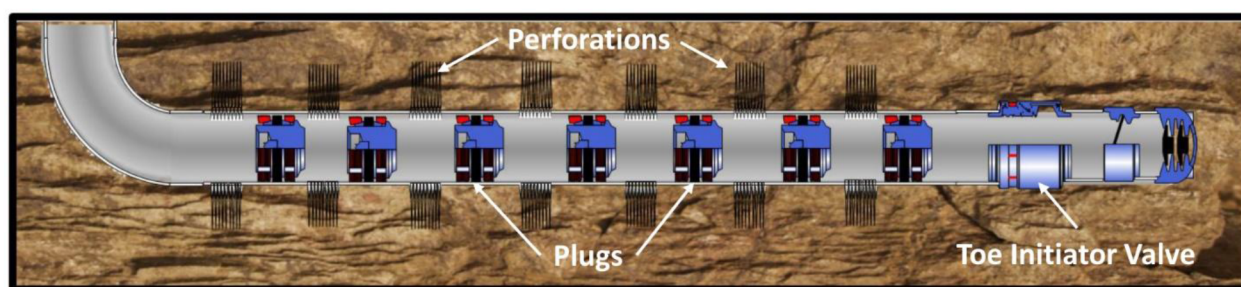


Figure 1—Example of MSF with TIV (et al. Rahim 2015)

OHMSF completions have OH packers between stages and those packers must be placed in accurate positions to ensure isolation between stages. Figure 2 below illustrates SE OHMSF completions.

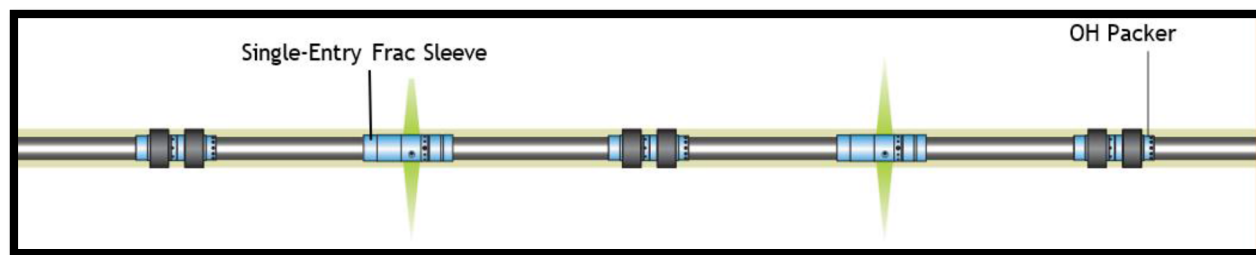


Figure 2—Example of SE OHMSF Completion (2 Stages)

If some OH packers are not in correct positions to isolate or they are not activated properly for any reason, there could be communication between stages during stimulation and certainly during production. Ball drops to gain stimulation efficiency become of no use as stages are not stimulated effectively or stages later on cannot be isolated properly if closing of stages is an option. This becomes a bigger issue in ME OHMSF (illustrated in Figure 3 below) as it is intended to stimulate more pay contact but if OH, constrictor packers, or constrictors (illustrated in Figure 4 below) are not utilized properly then we have an overdesign issue and ball drops for efficiency gain is not fully realized and just cost is added.

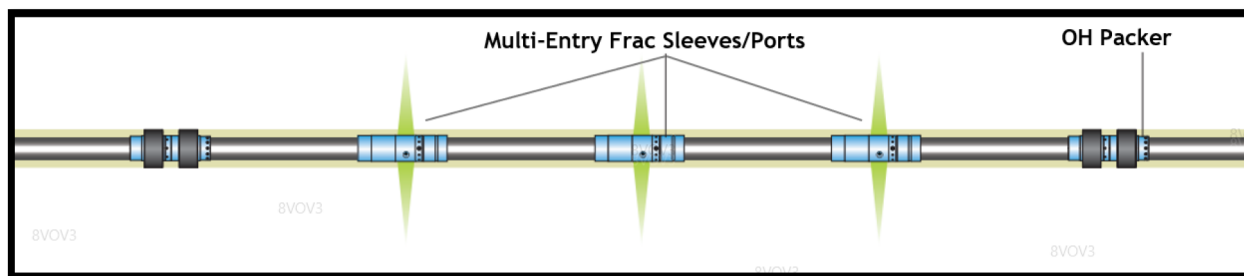


Figure 3—Example of ME OHMSF Completion (1 Stage)

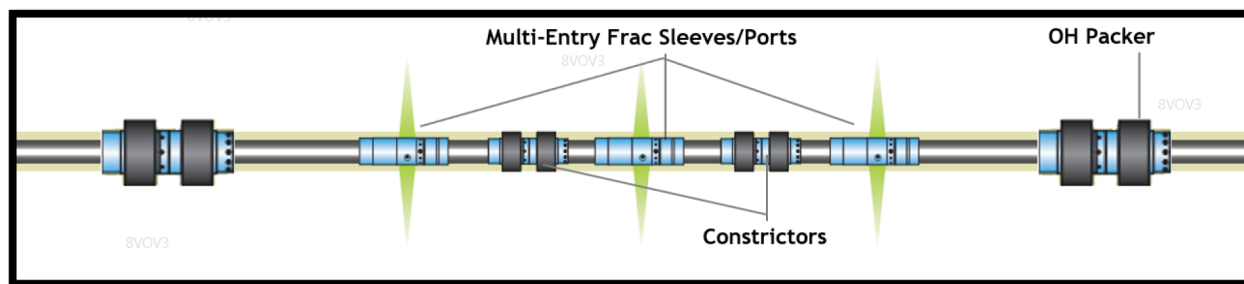


Figure 4—Example of ME OHMSF Completion with Constrictors (1 Stage)

Cemented Multi-Stage Fracturing (CMSF) Completions

CMSF completions came with the same concept of dropping balls to activate ports or fracturing sleeves post the TIV stage. Cement is used between stages or even within a stage between the fracturing sleeves as the constrictors in OHMSF completions. It was noted that cement between stages or in between fracturing sleeves within a stage could work better than OH packers between stages or constrictor packers within the same stage especially if an induced fracture is needed in a stage or in an entry within a stage to produce. Also, during the production phase, cement is expected to stay intact longer than conventional OH packers. New OH packers were introduced recently that are expected to stay longer throughout the life of a well or producing lateral but they are still costly to be used than cement.

Single-Entry (SE) Cemented Multi-Stage Fracturing (CMSF)

SE CMSF was inspired by the conventional SE OHMSF with OH packers but cement is used instead of packers as illustrated below in Figure 5. The use of cement can be more economical between fracturing sleeves/stages instead of potential costly OH packers. Ball seats are normally under the fracturing sleeves and when a ball is dropped, it activates a sleeve and then lands in a seat.

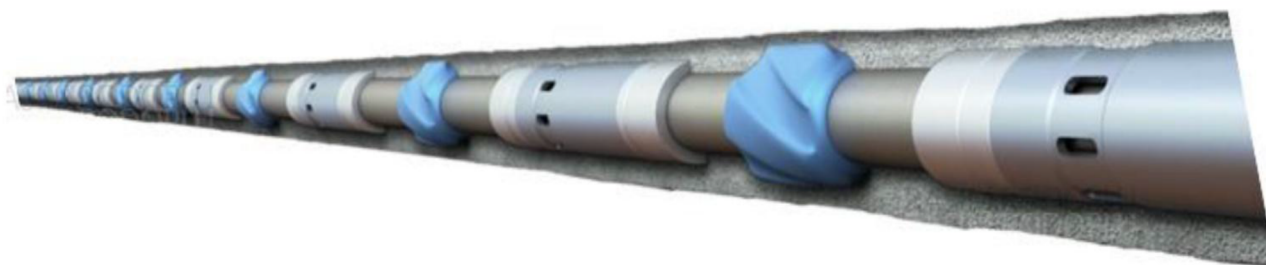


Figure 5—Example of SE CMSF Completion, 9 SE Stages (et. al Rodoplu 2023)

The first gas well that tested the SE CMSF had excellent production. It was planned with 9 stages including the TIV stage to be stimulated but only managed to stimulate 7 stages successfully. Time to

stimulate the stages were almost double the planned time as 2 stages had low injectivity and did not allow for continuous uninterrupted stimulation operation. HPCT was used multiple times to improve injectivity and mill balls and ball seats. The first gas well with SE CMSF did flow commercial gas rate but it was clear that operation efficiency during stimulation need to improve.

Multi-Entry (ME) Cemented Multi-Stage Fracturing (CMSF)

The ME CMSF completion was introduced to enhance the injection rates during stimulation and avoid interruptions during stimulation for faster well delivery. Also, it is used to maximize production as more entries are expected to be stimulated than SE CMSF completions and even potentially more effective entries for fracturing than the ME OHMSF. Figure 6 below illustrates the concept of ME CMSF systems where cement is used between stages like the SE CMSF and also between fracturing sleeves within a stage. Also, ball seats are normally below the sleeves for each stage. When a sized ball for a stage is dropped, it opens all fracturing sleeves for a specific stage and lands on the specific ball seat for that stage.



Figure 6—Example of ME CMSF Completion, 3 Entries or 1 Stage (et. al Rodoplu 2024)

The first gas well to utilize the ME CMSF was planned with 10 stages including the TIV stage. The TIV stage had two entries while the remaining stages had three entries per stage. This well could have used less entries than needed but it was determined that way to ensure smooth injection and stimulation days not to exceed the planned days. 10 stages were pumped in almost half the planned stimulation days showcasing that the ME systems are more environmentally efficient and the way to go especially in tighter gas developments. It was determined later on that some entries could have been dropped to avoid an overdesign depending on determined reservoir metrics with the latest reservoir engineering tools. ME CMSF provides more contact points with heterogeneous reservoirs, offering multiple entry points or pathways, which can enhance production and ensure more effective stimulation. Also, if some completions are not expected to run to bottom or the total depth of wells for any reason, having backup entries can be recommended.

ME CMSF (Well A) & SE CMSF (Well B)

The following key performance indicators (KPIs) were chosen to assess the performance of a ME CMSF system (Well A) over a SE CMSF system (Well B), total cost of the wells, delivery time and production gain. As gas reservoirs are extremely heterogeneous, production gains can be normalized to reservoir quality, i.e. k.h. As for the total cost or delivery time, it can be broken down for drilling, completion, and stimulation phases. If we are drilling in the same quality rocks/formations, we expect the drilling costs and time to be about the same. Completion costs are expected to be slightly higher for the ME CMSF system as more entries

or frac sleeves are used. However, stimulation costs are expected to be lower for ME CMSF completions driven by no interruptions during stimulation.

Total Cost Comparison

Well A managed a total cost about 2% less than Well B that had the SE CMSF system as illustrated in Figure 7 below.

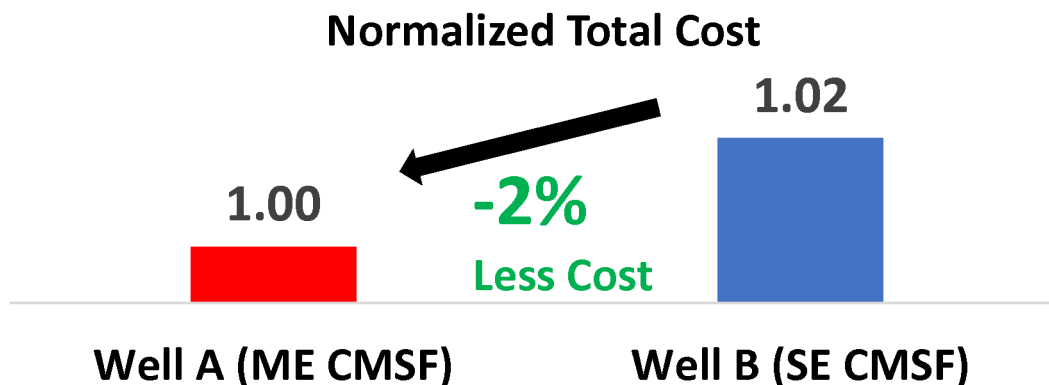


Figure 7—Total Cost Comparison (SE vs ME CMSF)

Stimulation Time

Well A with ME CMSF significantly stimulated with 75% less time compared to Well B with SE CMSF as illustrated in Figure 8 below. If Well B had more entries per stage in some stages or less total number of stages, it could have been pumped with in similar time to Well A. This requires comprehensive understanding of the targeted reservoir properties to avoid missed opportunities. In general, a good quality reservoir can have SE stages and a tighter reservoir should have ME stages to ensure uninterrupted stimulation.

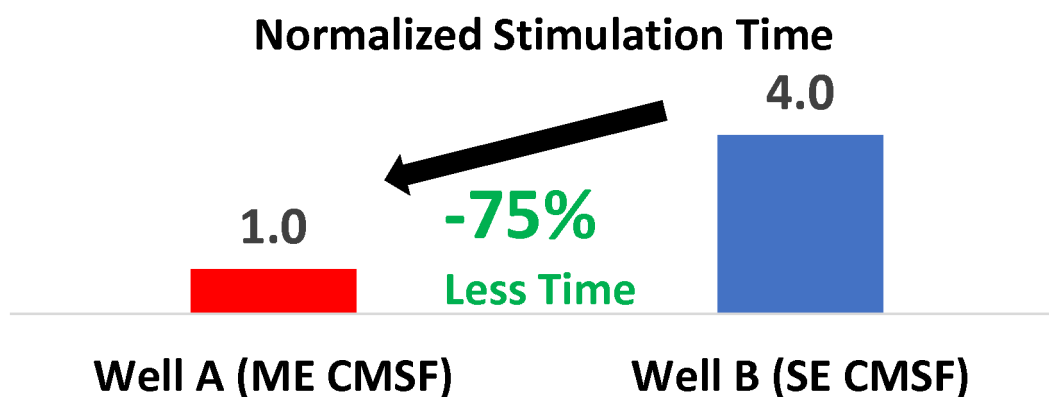


Figure 8—Stimulation Time Comparison (SE vs ME CMSF)

Production Gain

Both Well A and B did meet their production targets. However, normalized productivity index (PI) incorporating reservoir quality (k.h), show that Well A has significantly higher normalized PI by 2.7 folds as illustrated in Figure 9 below. Also, gas rates are picked at a common flowing pressure.

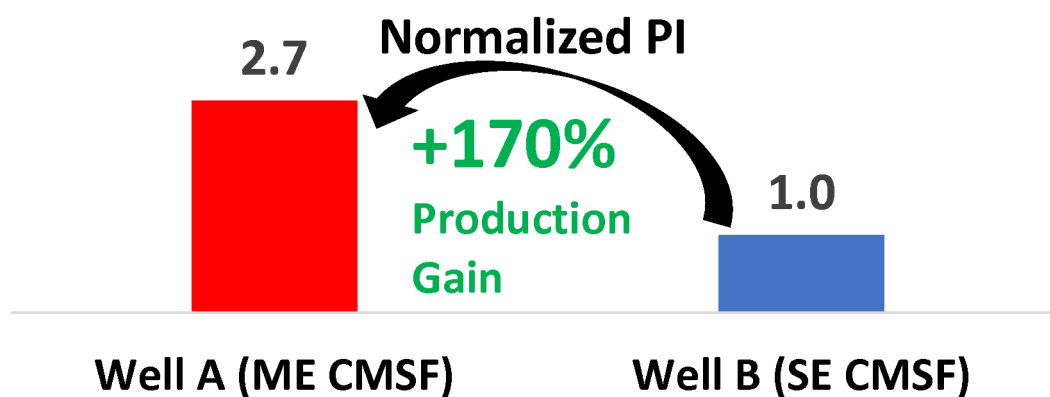


Figure 9—Normalized Productivity Index Comparison (ME vs SE CMSF)

Conclusion

- Choosing optimum number of stages and entries in MSF completions is extremely difficult but when integrated engineering is done, it can be eased and satisfactory results are met. It is coupled with choosing the right formation to drill into, following the best horizontal drilling practices, acquiring the needed logs, and choosing the latest advanced completions. ME CMSF completion seems to satisfy most stakeholders whether during planning, drilling, or during stimulation.
- It was noted here that cost went down by 2% but this percentage can certainly improve when looking in details about the cost of each phase when developing a gas producer.
- Stimulation time was dropped by 75% when comparing a specific SE well with another specific ME well. In a perfect case scenario when reservoir heterogeneity is fairly understood, a SE system or ME system stimulation time may be the same if no stimulation interruptions happen.
- ME CMSF is expected to deliver better productivity than SE CMSF especially when looking at a normalized PI respecting well pressure and reservoir quality (k.h).
- If ME CMSF is chosen in poor reservoir quality (or low k.h), it is not expected to deliver magic and poor results will be encountered.
- Finally, to find the optimum number of stages or entries even in conventional CL completions, a great knowledge of reservoir engineering is required. Future work is planned to show more details.

Nomenclature

AI	: Artificial Intelligence
CL	: Cemented Liner
HPCT	: High-Pressure Coiled Tubing
k.h	: Permeability-Thickness Product
KPI	: Key Performance Indicator
LWD	: Logging While Drilling
ME	: Multi-Entry
MSF	: Multi-Stage Fracturing
OH	: Open-hole
P&P	: Plug and Perf
PI	: Productivity Index
SE	: Single-Entry
TIV	: Toe Initiator Valve

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